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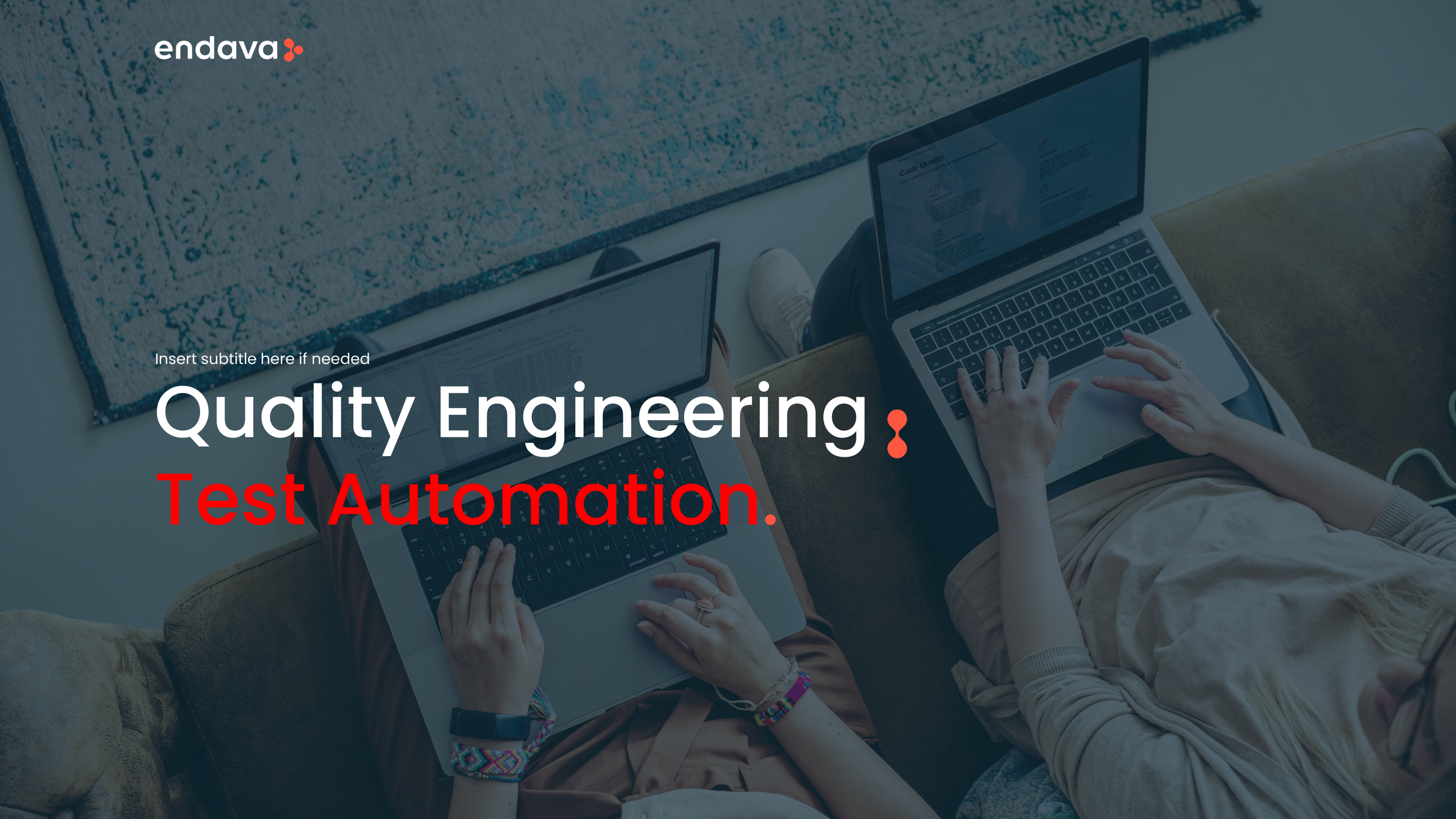
Insert subtitle here if needed

Quality Engineering: Test Automation.



Agenda

01 History

02 Why test automation?

03 What to automate?

04 How AI expands the scope

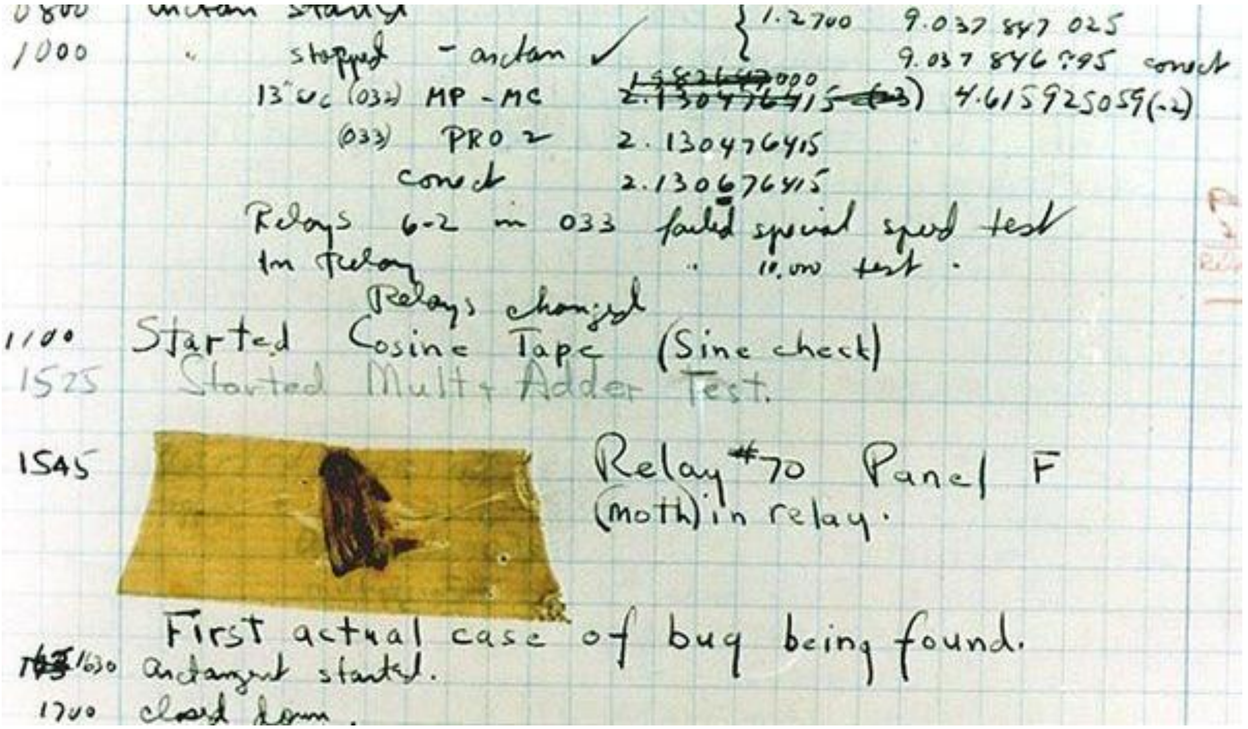
05 The AI-Enhanced Automated Testing Process

06 AI-Driven Test Automation

07 Essential AI tools and libraries for Analysis and testing

08 Q&A

First actual case of “Bug” being found.



Chapter

01 History

40’s Manual Testing (Programmers)

70’s Manual Testing (Testers)

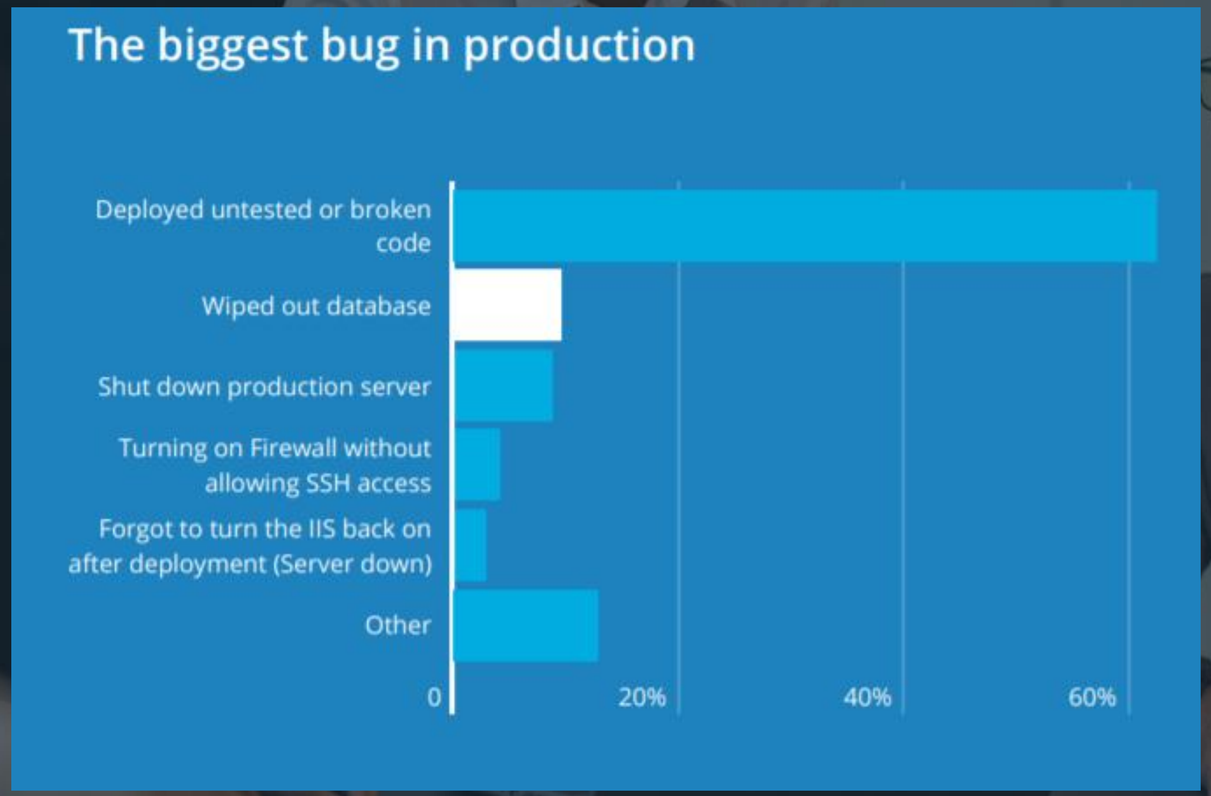
80’s Capture/Replay Testing

90’s Script-Based Testing

Today – AI Spec Driven Testing

Chapter

02 Why to automate?



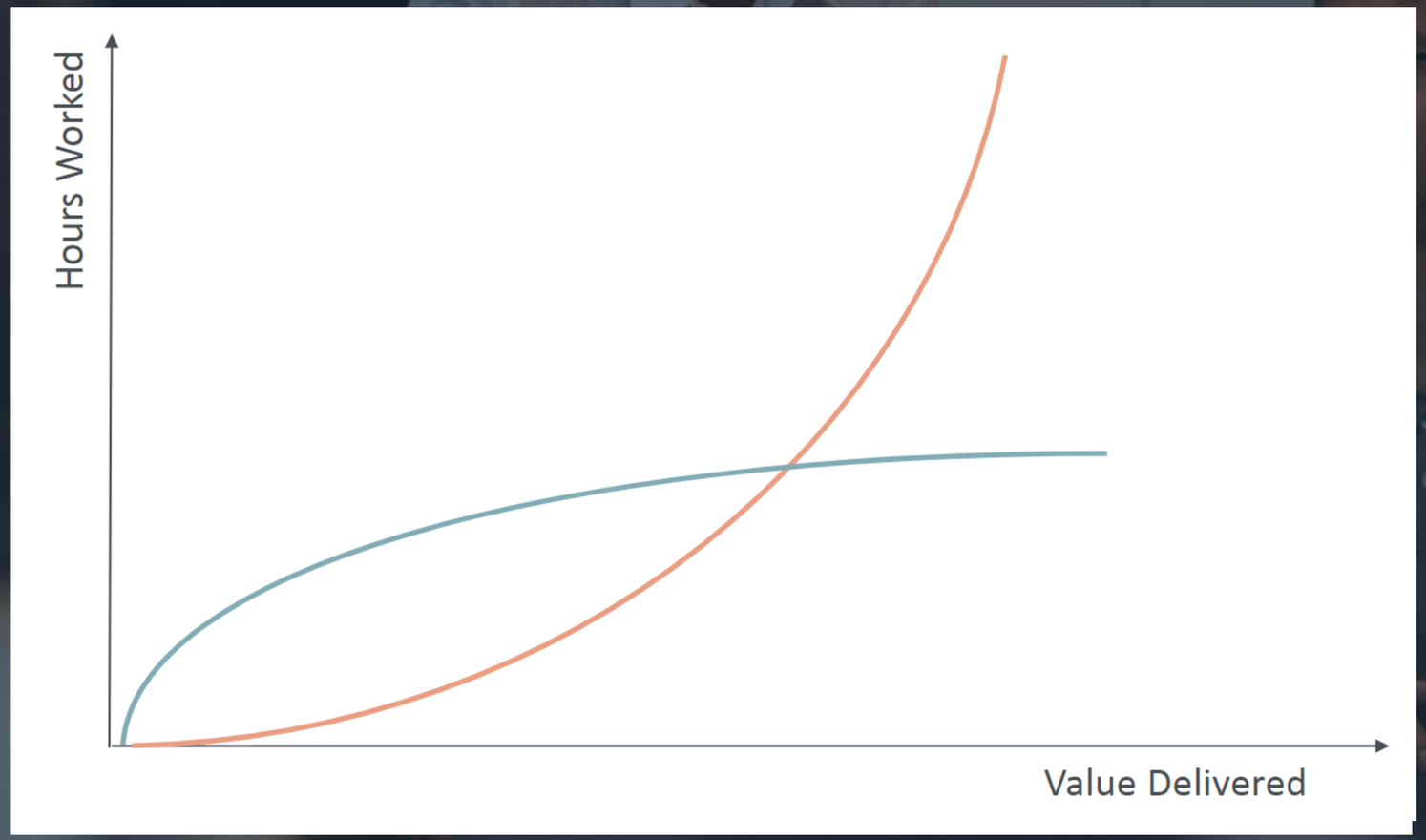
Chapter

02 Manual vs Automation

	Manual Testing ✅ ❌	Testing Automatizado ✅ ❌
Tiempo de ejecución	Lento ❌	Rápido ✅
Coste inicial	Bajo ✅	Alto ❌
Mantenimiento	Bajo ✅	Alto ❌
Exploración visual	Excelente ✅	Limitado ❌
Repetibilidad	Baja ❌	Alta ✅
Integración en CI/CD	No ❌	Sí ✅

Chapter

02 Manual vs Automation



Why should we automate?

- Reduction of repetitive processes
- Saves time and money
- Increases test coverage
- Manual testing can become error prone
- Less human intervention
- Quick feedback
- Scheduled execution
- Report results

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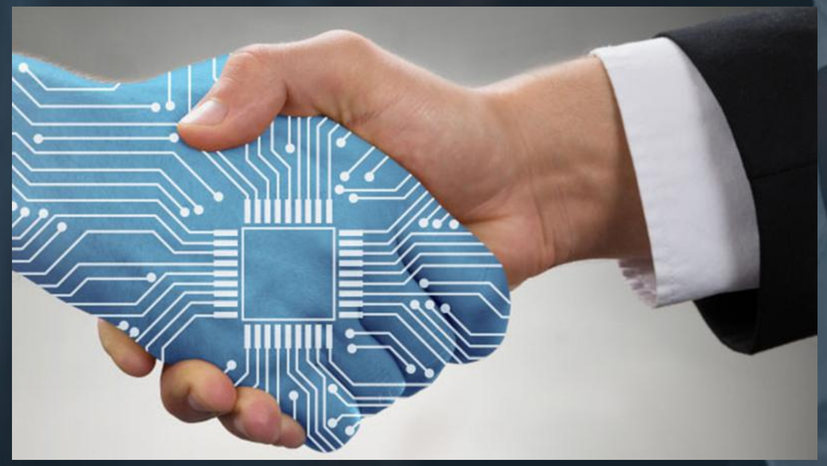
Chapter

03 What to automate?

Traditional approach:



- High Risk – business critical flows
- Stable Flows
- Repetitive Flows
- Very tedious or difficult to perform manually
- Time consuming scenarios

Test cases
We should not

automate.

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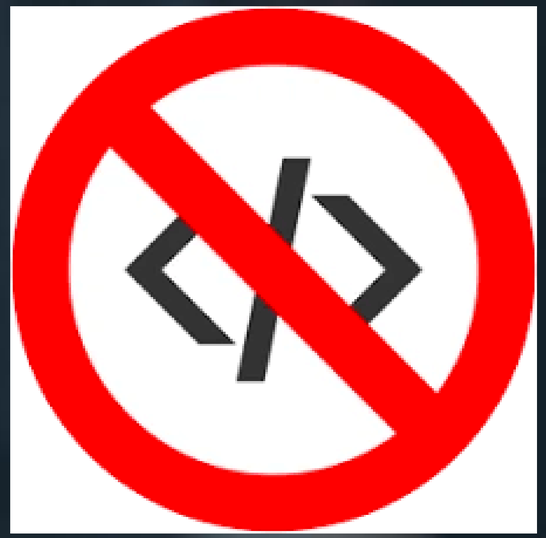
Chapter

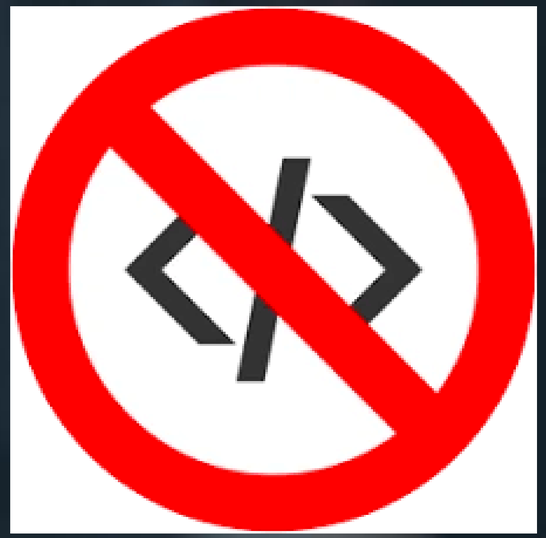
03 Not suitable for automation



- Newly designed test cases which hasn’t been executed manually at least once.
- When the requirements for that test case change frequently.
- Test cases which are executed on ad-hoc basis (usability, accessibility, exploratory).

Chapter

03 Limitations

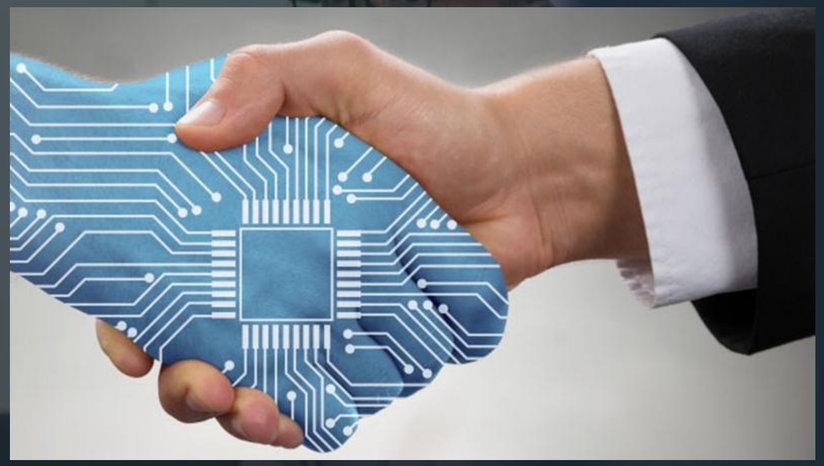


- Two factor authentication
- reCAPTCHA
- Difficult to test email text content
- Mobile sensor emulation or reading with camera
- UX
- Exploratory testing

04. How AI Expands the Scope

AI expands the scope

- Auto-Healing Locators
- Intelligent Test Data Generation
- Intelligent Test Suite Execution (Impact-Based Testing)
- Automated Root Cause Analysis
- Semi-Automated Exploratory Testing



Auto-Healing Locators

- Alternative attributes are analyzed
- Previous DOM structure is compared
- Similar elements are identified by text, hierarchy, or position
- The locator is automatically repaired

JavaScript

```javascript
await page.locator('#submit-btn').click();
```

JavaScript

```javascript
await page.getByRole('button', { name: /submit/i }).click();
```

Chapter

Test Data Generation

- Dynamic data generation
- Realistic user simulation
- Broader input combinations
- Detect invalid input patterns
- Automatically generate edge cases
- Create intelligent negative scenrios

JavaScript

```javascript
await page.getByLabel('Email').fill('test123@test.com');
```

```javascript
import { faker } from '@faker-js/faker';

test('User registration with dynamic data', async ({ page }) => {
  const email = faker.internet.email();

  await page.goto('/register');
  await page.getByLabel('Email').fill(email);
  await page.getByLabel('Password').fill('Password123');
  await page.getByRole('button', { name: 'Register' }).click();

  await expect(page.getByText('Welcome')).toBeVisible();
});
```

Intelligent Test Suite Execution

```
npx playwright test
```

```
npx playwright test tests/checkout.spec.js
```

With AI-enhanced CI/CD you could:

- Analize which files changed
- Run only impacted tests
- Prioritize tests by risk

This expands automation since you are testing execution strategy.

- Automatic commit analysis
- Dependency mapping
- Historical failure patterns

Automated Root Cause Analysis

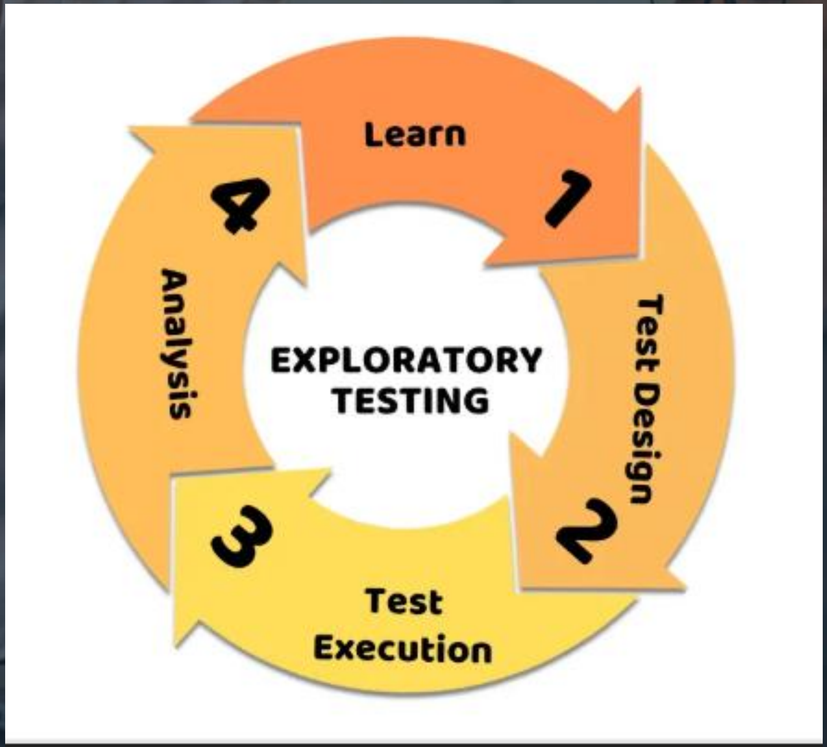
- Analyze stack traces
- Detect recurring patterns
- Classify flaky tests
- Identify backend latency issues

```
Timeout 30000ms exceeded while waiting for element
```

Chapter

Semi-Automated Exploratory Testing

- Analyze stack traces
- Detect recurring patterns
- Classify flaky tests
- Identify backend latency issues



05. The AI-Enhanced Automated Testing Process

Intelligent Execution

We are no longer just automating test scripts, we're automating:

- Execution decisions
- Failure analysis
- Risk prediction
- Environment preparation

The testing process becomes intelligent and adaptative, not just automated.

Dynamic Test Suite Optimization

When we run the entire suite:

- Slow feedback
- Unnecessary execution
- Wasted CI resources
- Delayed releases

AI-Enhanced Approach:

AI analyses:

- Git commit changes
- Impacted files
- Code coverage
- Historical failure
- Risk areas

Instead of running all tests every time, AI decides which tests matter based on what changed.

Intelligent Reporting

❖ AI-Driven Root Cause Analysis like:

- Logs
- Network traces
- Screenshot changes
- Historical failure patterns

Then it is classified:

- Backend timeout
- Selector issue
- Flaky test
- Infrastructure problem

Instead of just showing that the test failed, AI explains why it likely failed.

06. AI-Driven Test Automation

Chapter

AI-Driven Test Automation and Debugging Techniques

AI in Test Automation Creation and Maintenance

- Intelligent Test Case Generation
- Self-Healing / Adaptive Locators
- Intelligent Test Prioritization
- Test data optimization

AI in Debugging and Failure Analysis

- Automated Root cause Classification
- Flaky Test Detection
- Failure Clustering
- Predicting Debugging

Intelligent Test Case Generation

Traditionally:

- Requirements
- User stories
- Acceptance Criteria

With AI:

- Requirements can be parsed automatically
- APIs can be analyzed automatically
- UI flows can be inferred
- Edge cases can be generated

Example:

Given this requirement:

- "User must enter a valid email and password to login"

The AI can generate:

- Valid email, valid password
- Invalid email formats
- SQL injection attempts
- Boundary-length passwords
- Unicode characters

Instead of writing 5 test cases manually, AI can generate 50 meaninfull variations

Self-Healing / Adaptative Locators

Traditionally:

- Requirements
- User stories
- Acceptance Criteria

With AI:

- Requirements can be parsed automatically
- APIs can be analyzed automatically
- UI flows can be inferred
- Edge cases can be generated

07. Essential AI tools and Libraries for Testing

ML Libraries for Custom Models

What is a ML library?

- Learn from data
- Detect Patterns
- Make predictions
- Classify information

What are they used for in testing?

- Detect patterns of recurring failures
- Predict which tests are more likely to fail
- Detect flaky tests
- Automatically classify erros
- Detect anomalies in logs
- Optimize test suite execution

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Most important libraries

Tool	Best For	Key AI Feature	My Connection
BlinqIO	Cucumber + GenAI	AI meets prompt engineering	Had founders on podcast A485
testers.ai	Autonomous everything	AI agents write & run tests	Ex-Google Chrome testing team
Mabl	Agentic workflows	Autonomous test agents	Multiple podcast appearances
Katalon	All-in-one platform	Self-healing + AI generation	Gartner Magic Quadrant pick
Applitools	Visual testing	Visual AI pioneer	Adam Carmi on A450
ACCELQ	Codeless automation	Generative AI creation	Guild sponsor + webinars
BrowserStack	Test observability	AI root cause analysis	Long-time Guild partner
Testim	Reducing flaky tests	ML-powered locators	Oren Rubin interview
LambdaTest KaneAI	LLM-powered	Natural language tests	Cloud testing platform
TestResults.io	No selectors	Selector-free testing	Tobias on podcast
Tricentis	Enterprise	Fully codeless AI	Guild conference sponsor
Parasoft	Manual regression testing	Code coverage-driven test prioritization	Wilhelm & Daniel demo

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Data Processing Tools for log and telemetry analysis

1. Full-Stack Observability Platforms

Unifican logs, métricas y trazas para una visibilidad completa del sistema.

Category	Tool	Key Strength	Best For
Full-Stack Observability	**Datadog**	Unified logs, metrics, and traces with automated correlation.	E2E testing in complex cloud environments.
Full-Stack Observability	**New Relic**	"Logs-in-context" and powerful NRQL querying.	Correlating app performance with test failures.
Full-Stack Observability	**Dynatrace**	AI-driven root cause analysis (Davis AI).	Large-scale enterprise systems and microservices.

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Data Processing Tools for log and telemetry analysis

2. Log Management & Centralization

Especializadas en la búsqueda, almacenamiento y análisis de grandes volúmenes de registros.

Category	Tool	Key Strength	Best For
Log Management	**ELK Stack**	Highly flexible search (Elasticsearch) and visualization (Kibana).	Teams needing a customizable, industry-standard stack.
Log Management	**Splunk**	Advanced SPL searching and ML-based predictive analytics.	Large enterprises with high security/compliance needs.
Log Management	**Grafana Loki**	Cost-efficient log indexing, optimized for Kubernetes.	Cloud-native teams using Grafana dashboards.

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Data Processing Tools for log and telemetry analysis

3. Specialized Test Reporting Tools

Herramientas diseñadas específicamente para visualizar resultados de automatización.

Category	Tool	Key Strength	Best For
Test Reporting	**ReportPortal**	AI-powered failure classification and historical analysis.	Categorizing flaky tests and identifying bug patterns.
Test Reporting	**Allure Report**	Rich visual reports with embedded logs and screenshots.	Quick integration with Selenium, Playwright, or Cypress.

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Data Processing Tools for log and telemetry analysis

4. Data Collection & Pipelines

Estándares y agentes para recolectar y transportar telemetría.

Category	Tool	Key Strength	Best For
Data Collection	OpenTelemetry	Vendor-agnostic standard for traces and metrics.	Future-proofing instrumentation across different tools.
Data Collection	Fluent Bit	Lightweight log processor and forwarder.	Routing telemetry from containers with low overhead.

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08.

Q & A

Thank you!